

SMART UTM — PROGRESS REPORT

Jönköping University · School of Engineering — status to 12 August 2026 · automated tensile testing with camera-based strain

12 of 16 planned features are built AND validated on the rig.

All six closed-loop test protocols work. The material results are publication-grade. What remains is two rig runs, three convenience features, and two hardware decisions that are not mine to make.

12 / 16 features built and rig-validated	6 / 6 test protocols validated	13 specimens tested, 17 indexed runs	2.5 % strength scatter, n = 5 at 100 % infill	2 hardware decisions needed from you
---	---	---	--	---

WHERE THE PROJECT STANDS

The rig is no longer a manual machine. A test is now: pick a recipe, press Prepare, press Start — the software runs the protocol, watches the specimen through a camera, stops itself at fracture and writes an indexed report. Six protocols beyond a plain pull are validated, so a single specimen can return a curve rather than a single number. The two results that were negative in July, cyclic hysteresis and creep, were both re-sized and are now positive.

The rig — built in-house, every DIC control loop closed in software



Two steppers (1.85 Nm, 20:1) drive 5 mm lead screws against a fixed base. An ANYLOAD 3 t cell measures force; a Basler camera measures strain optically at 20.8 px/mm from two markers on the 80 mm gauge. The ESP32 runs STEP/DIR/ENABLE only and knows nothing of test protocols — every protocol, guard and measurement here lives in the PyQt6 app, closing the loop at ~11 Hz.

WHAT HAS BEEN DELIVERED

- June** • **DIC validation closed**
41/41 software checks; tension, compression and staircase linearity all sign-verified.
- June** • **First run to fracture**
Rig and camera both hold all the way to failure — the precondition for every destructive protocol since.
- June** • **100 % infill quintet, n = 5**
UTS 46.2 ± 1.2 MPa, scatter 2.5 % — publication-grade repeatability.
- July** • **Smart features SF1–SF10**
One-click workflow, a four-layer safety net, and constant-strain-rate control demonstrated to fracture.
- August** • **All six protocols validated**
Including both destructive protocols; two landed 0.9 % apart on ultimate strength, though on different specimens.
- August** • **Fatigue and creep resolved**
Residual strength after cycling is –6.8 % against the same-protocol virgin run; creep resolved at 41× the noise floor.

RESULTS WORTH REPORTING

Strength repeatability is publication-grade

100 % infill, n = 5: UTS 46.2 ± 1.2 MPa — 2.5 % scatter, and the only n > 2 result we own. Two fracture protocols landed 0.9 % apart, but on DIFFERENT specimens (n = 1 each), so that bounds protocol and specimen scatter together, not the protocol alone.

The camera is not a nicety — it changes the answer

Over the same cycles the crosshead reads 4 % STIFFER while the specimen's true modulus falls 26 %. Without strain measured on the specimen itself, this test reports the opposite of what actually happened.

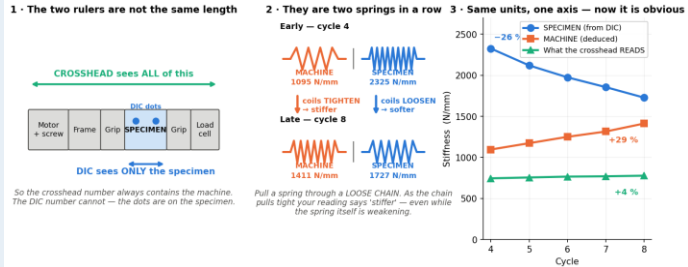
First fatigue-damage number

After 16 cycles at 79 % of fracture load, residual strength is –6.8 % against the same-protocol virgin run (–7.2 % against the two-protocol mean). The virgin baseline spans only 0.9 %, so the loss is far larger than the scatter.

Creep measured, with the instrument subtracted

A zero-load baseline measured the rig's own drift first; the corrected creep is +1080 $\mu\epsilon$ at 41× the noise floor, and decelerates as real creep must.

The machine reads STIFFER while the specimen SOFTENS



Progressive cyclic to fracture. Crosshead stiffness ROSE 744 → 777 N/mm as rig slack squeezed out, while the DIC-measured specimen modulus FELL 2.32 → 1.73 GPa, a 26 % loss. Read from the crosshead alone this test says the material is getting stiffer — the strongest argument for putting the camera in the loop.

DECISIONS I NEED FROM YOU

Motor torque ceiling — needs a hardware decision

Fracture loads reach 3.2–3.4 kN normally, but fell to ~2.6 kN on one session and stalled a 100 % infill test. The drivetrain is rated above 12 kN, so the cause is electrical: driver current, thermal derating or supply sag. Two routes — set the driver current (Vref) on the board, or wire the TMC2160's SPI pins to the ESP32 so current and thermal status can be set and logged in software. I need bench time with the driver board, and a decision on which route.

Poisson's ratio — needs optics, not code

The four-marker maths is written and self-tested, but the gauge is too narrow to resolve a transverse pair at the present 20.8 px/mm. A gauge-zoomed lens (or a second camera) plus a matte backdrop unblocks true stress and Poisson's ratio. Until then every stress reported is engineering stress, which is stated on every figure.

More specimens before publication

Creep and fatigue are each n = 1 at one condition. The measurement is proven; the material constants are not. 3–4 stress levels for creep and 3 repeats for fatigue would make both publishable.

NEXT

Now T6.6 — the fatigue damage curve

One fresh specimen, 12 cycles, one uninterrupted run. The only missing piece of the damage story; about 40 minutes of rig time.

Now Black-specimen DIC check

Every specimen so far is white. The black preset has never been exercised — needed before anyone else uses the rig.

Next SF11 — auto-metadata and per-specimen reports

Removes the last manual bookkeeping step from a test.

Next Student operating manual

So the rig can be run by someone who did not build it.